

# Jerome Thayananthajothy

Technical Lead | AI-Driven Systems Architect | AWS & Full-Stack Expert | Open Source Contributor

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## Professional Summary

I'm a Technical Lead and Generative AI Engineer focused on building intelligent, high-scale systems that merge traditional full-stack engineering with next-generation AI capabilities. Over the past decade, I've led architecture modernization initiatives across e-commerce, analytics, and workflow automation driving multimillion-pound cost reductions, boosting performance, and introducing AI-augmented features like semantic search and agentic workflows. I'm particularly interested in AI-enhanced infrastructure, multi-cloud orchestration (AWS, Azure, GCP), and scalable developer ecosystems. My engineering philosophy is rooted in system design, observability, and enabling teams to ship fast with confidence.

## Core Expertise

**Full Stack:** SOC, IAM, Search Engines, PostgreSQL, MuleSoft Anypoint Platform, Scrum, Agile Methodologies, MongoDB, Software Documentation, Web Applications, Managing Technical Personnel, Node.js, Inertia.js, React.js, .NET Framework, C#, Jira, GraphQL, Eloquent, RestAPI, Databases, SQL, Query, Cybersecurity, TypeScript, VueJS, GitHub, JavaScript Libraries, AJAX, JSON, JavaScript Frameworks, Mocha (JavaScript Framework), Jest, JavaScriptMVC, Vanilla JavaScript, Server Side JavaScript, MySQL, Object-Oriented Programming (OOP), PHP, JavaScript, Python, Algorithms, Computer Networking, Adobe Photoshop, Cascading Style Sheets (CSS), WordPress, jQuery, HTML, Bootstrap, HTML5, Laravel, Django, Vue.js, PhpStorm, PHP Applications, PHPUnit, PHP Frameworks, Emberfuse, Java, C++, C, Go, JavaSE, Vue, Vuex, Nuxt.js, Symfony

**AI & Architecture:** Artificial Intelligence (AI), GenAI, Microservices

**Cloud & Delivery:** Microsoft Azure, AWS, Amazon Web Services (AWS), Cloud Computing

**Leadership:** Technical Staff Management

## Experience

**Technical Lead** May 2026 – Present

*Sino Lanka Group* — Colombo, Western Province, Sri Lanka

- Guides architecture direction across full-stack systems and cloud delivery.
- Focuses on reliable engineering practices and observability.

**Technical Lead** Oct 2025 – May 2026

*KAYA* — Colombo

- Designed AI workflow engines with LangGraph.
- Led distributed application work across Kubernetes clusters.
- Worked across Node.js, TypeScript, Laravel, and PostgreSQL.

**Technical Lead** Dec 2024 – Sep 2025

*Brompton Bicycle* — Greenford, England, United Kingdom

- Built an AI-powered full-text search engine, augmenting MySQL's native capabilities with semantic and vector-based search for product discovery and analytics.
- Integrated and orchestrated multi-stack systems, including MuleSoft for service orchestration and Oracle NetSuite for ERP synchronization and order lifecycle automation.
- Reduced checkout latency from 4.2s -> 1.5s (60%+ improvement), boosting conversion rates by 12%.
- Shifted release cadence from monthly to biweekly and reduced technical debt by 30%, enforcing new internal architecture and CI/CD standards.

**Senior Software Engineer** Jul 2023 – Dec 2024

*NetWatch Global®* — Birmingham, England, United Kingdom

- Established and enforced industry standards for the engineering department and all company software, contributing to a more efficient and cohesive development process.
- Worked extensively with PHP, TypeScript, VueJS, AWS DynamoDB, and MySQL/PostgreSQL, leveraging these technologies to deliver robust and scalable solutions.

**Early Career** 2015–2022

Senior Software Engineer, MacroActive™ (2021–2022); Senior Software Engineer, Elegant Media Australia (2020); Software Engineer, Elegant Media Australia (2019–2020); Software Engineer, Seven's Idea (PVT) Ltd (2018–2019); Software Engineer, Alta Vision Solar (2017–2018); Telecommunications Engineering Intern, Sri Lanka Telecom (2016); Telecommunications Engineering Intern, Sri Lanka Telecom (2015–2016); Associate Software Engineer, Alta Vision Solar (2015–2017).

## Selected Open-Source Work

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|                                                                                                                                                            |                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Fetch PHP</b>                                                                                                                                           | 450 stars   PHP        |
| Modern PHP HTTP client inspired by JavaScript's fetch API. Async/await, streaming, SSE, middleware, RFC 7234 caching, and full PSR-7/18 compliance.        |                        |
| <b>Filterable</b>                                                                                                                                          | 194 stars   PHP        |
| Enhance Laravel queries with adaptable, customisable filters and intelligent caching to improve both performance and functionality.                        |                        |
| <b>phpvm</b>                                                                                                                                               | 171 stars   Shell      |
| Simple PHP version manager for MacOS and Linux.                                                                                                            |                        |
| <b>Comet</b>                                                                                                                                               | 157 stars   TypeScript |
| A cross-platform desktop app powered by FFmpeg for easy, beautiful media conversion.                                                                       |                        |
| <b>Matrix</b>                                                                                                                                              | 60 stars   PHP         |
| Bringing JavaScript-style async to PHP! Manage asynchronous tasks with ease using a modern, fiber-powered API for seamless concurrency and error handling. |                        |

## Product Work

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| <b>Euclidius</b>                                                                                                                                                                                      | Creator and builder |
| A hate speech detection neural network which analyzes sentence sentiment to determine if the content is hateful, offensive, or acceptable.                                                            |                     |
| <ul style="list-style-type: none"> <li>Analyzes sentence sentiment for content moderation.</li> <li>Classifies text as hateful, offensive, or acceptable.</li> </ul>                                  |                     |
| <b>Laravel Filterable</b>                                                                                                                                                                             | Creator and builder |
| Enhance Laravel queries with adaptable, customisable filters and intelligent caching to improve both performance and functionality.                                                                   |                     |
| <ul style="list-style-type: none"> <li>Provides reusable filters for Laravel queries.</li> <li>Uses intelligent caching to improve query performance.</li> </ul>                                      |                     |
| <b>Hands of Mercy – Charity Platform</b>                                                                                                                                                              | Creator and builder |
| Laravel + Vue system enabling sponsorships and donation management.                                                                                                                                   |                     |
| <ul style="list-style-type: none"> <li>Supports sponsorship workflows.</li> <li>Handles donation management in a Laravel and Vue application.</li> </ul>                                              |                     |
| <b>Altus 4 – AI-Enhanced MySQL Search Engine</b>                                                                                                                                                      | Creator and builder |
| Search-as-a-Service platform augmenting MySQL full-text with semantic search, query optimization, and analytics.                                                                                      |                     |
| <ul style="list-style-type: none"> <li>Combines MySQL full-text search with semantic search.</li> <li>Includes query optimization and search analytics.</li> </ul>                                    |                     |
| <b>Cerberus IAM – Identity &amp; Access Management Platform</b>                                                                                                                                       | Creator and builder |
| PERN-based IAM solution with dynamic organization directories, SSO, and admin SDK.                                                                                                                    |                     |
| <ul style="list-style-type: none"> <li>Supports dynamic organization directories and single sign-on.</li> <li>Provides an admin SDK for identity and access management integrations.</li> </ul>       |                     |
| <b>Honeymelon</b>                                                                                                                                                                                     | Founder and builder |
| A premium Apple Silicon macOS app for fast, reliable media conversion, built as a focused commercial product.                                                                                         |                     |
| <ul style="list-style-type: none"> <li>Product-led work covering UX, packaging, performance, and release quality.</li> <li>Connects engineering craft with market-facing product judgment.</li> </ul> |                     |

## Education

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|---------------------------------|---------------------|
| <b>Bachelor of Science - BS</b> | Sep 2022 – Jul 2023 |
| <i>University of Sunderland</i> |                     |
| <b>Higher National Diploma</b>  | Aug 2020 – Mar 2022 |
| <i>ICBT Campus</i>              |                     |

## Certifications

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| AWS Certified Solutions Architect — Amazon Web Services (AWS)   Amazon Web Services Cloud Practitioner — Amazon Web Services (AWS)   Associate Member — Institution of Engineers Sri Lanka |  |
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## Languages

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| English (Native or bilingual proficiency)   Sinhala (Limited working proficiency)   Tamil (Native or bilingual proficiency) |  |
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